



University of St Andrews

Ethics Application Form

Introduction

Ethics Application

What type of form do you wish to complete?

New ethics application - select this for any new application for ethics review of research involving humans. This includes applications to renew a favourable opinion which is due to expire and ethics applications which will be accompanied by an SSER. Please check the [guidance on what projects require ethics review](#).

Ethics amendment application – select this only after using the 'request amendment' function. Do not use this for newly created projects.

Ratification of an external ethics favourable opinion/approval - select this if you already have an ethics favourable opinion or approval from an external organisation. This will prompt you to upload your external documents and route you past parts of the form. Please check the [guidance on projects already reviewed by another organisation](#).

None of the above – select this if you are completing another form with no ethics application, for example a standalone SSER with no ethics review application.

(SSER – Security-sensitive and extremism-related research declaration, see the [guidance on 'additional documents'](#))

- ☒ New application (including renewals)
- ☐ Amendment application
- ☐ Ratification
- ☐ None of the above

Is this application for research which has already started i.e. a retrospective application. For example, if you have already:

- completed the research
- started collecting human participant data or samples
- begun recruiting participants.

Please select 'no' to this question if you already have an ethics favourable opinion for the research which is due to expire and are seeking a renewal.

☐ Yes

☒ No

Security Sensitive Research Declaration

Are you completing a Security Sensitive Research Declaration?

☐ Yes

☒ No

Applicant Type

Applicant Details

Title

First Name

Surname

Email

Select applicant type (for the lead applicant)

- ☒ Undergraduate
- ☐ Staff
- ☐ PGR
- ☐ PGT
- ☐ Module Co-ordinator (for teaching module applications)
- ☐ Autonomous / non-affiliated project

Please add your supervisor.

Note: if your supervisor does not appear in the list, please ask them to visit this webpage ([Ethics Apply](#)) and log in. Once they have done that they will be visible in the list.

Title

Dr

First Name

Abd Alsattar

Surname

Ardati

Email

aaa8@st-andrews.ac.uk

Select the School ethics committee (SEC) which will be reviewing your application. This is usually the School of the lead applicant but, if you are a student, may be the School of your supervisor (check with your supervisor if you are unsure).

Computer Science

You have selected this School/ethics committee:

Computer Science

Please ensure this is correct before proceeding. An incorrect response may delay review of your application.

Research Type

Does the research involve fieldwork and travel?

- ☒ Yes
☐ No

Upload relevant documents such as risk assessments. Note: these are not part of the ethics review process but you can attach these if you have them or if your School ethics committee has requested copies (optional)

Does the project involve research intended to investigate, prevent or treat any disease, physiological or psychological condition or function?

[Click here for more information if needed.](#)

- ☐ Yes
☒ No

Does the research fit into either of these categories?

- Studies terrorism, extremism, terrorist or extremist organisations for groups, extremist ideologies, radicalisation/de-radicalisation
- Involves visits to websites that might be associated with radicalisation or terrorist/extremist organisations, groups or individuals

- ☐ Yes
☒ No

Does this project already have external ethics approval from another institution or organisation?

- ☐ Yes
☒ No

Is this application to renew an existing University ethics favourable opinion or approval i.e. one which is due to expire?

- ☐ Yes
☒ No

Is this application for a multi-phase/linked project i.e. related to another application

- ☐ Yes
- ☒ No

Data Type

What type of data/samples will you be using for this research?

For the avoidance of doubt, collection of data from individuals' social media accounts are considered primary data.

- ☒ Primary – data or samples collected or collated by the researcher for their research purpose
- ☐ Secondary - data or samples already collected or collated in a completed research project or accessed via a published research dataset/ repository / tissue bank
- ☐ Library, archival, or official media data – formally published or recorded material, such as articles, books, press releases, historical records.

Does the research involve use of human tissue or samples?

- ☐ Yes
- ☒ No

Participants

Please complete the filter questions - these determine whether your application will undergo standard review or proportionate review by your School ethics committee. If you are unsure which responses to select, please contact your [School ethics committee](#). For more information on the review process, visit the [Ethical review application](#) webpage.

Will your research involve the data or samples of children

- ☒ Yes
- ☐ No

Have you contacted the child panel representative for advice?

The child panel representative is Dr Barbara Dritchsel, School of Psychology and Neuroscience. Email: bd9@st-andrews.ac.uk

You need to contact the child panel rep for advice if your research involves children unless you are from the School of Medicine, School of Geography and Sustainable Development, or School of Psychology & Neuroscience.

- ☒ Yes
- ☐ No
- ☐ Not required

Will your research involve the data or samples of protected adults?

'Protected adult', in Scotland, means a person 16 years of age or older who, as a result of their particular needs (arising from a physical or mental illness, or a disability, which may disadvantage them), is in need of a registered care service, health service, community care service, or welfare service.

- ☐ Yes
- ☒ No

Will your research involve the data or samples of NHS patients or staff?

- ☐ Yes
- ☒ No

Will your research involve the data or samples of individuals engaged in criminal activity?

- ☐ Yes
- ☒ No

Will your research involve the data or samples of individuals in custody, care homes, or other residential institutions?

- ☐ Yes
- ☒ No

Will your research involve the data or samples of individuals impacted by a traumatic event such as war, displacement, acts of terrorism, abuse, discrimination, crime, disasters, life-changing illness or injury, bereavement?

- ☐ Yes
- ☒ No

Will your research involve the data or samples of individuals where there is any doubt over their capacity for freely given consent such as through cognitive impairment, language barriers, legal status, terminal illness?

- ☐ Yes
- ☒ No

Will your research involve the data or samples of any other individuals where the researcher or SEC identifies a vulnerability that cannot be satisfactorily mitigated?

- ☐ Yes
- ☒ No

Do you need PVG or DBS approval?

You may need this if your research involves unsupervised contact with children or protected adults whether in the UK or not. You can get advice on PVG and DBS, including how to apply, from:

- Students - Advice and Support Centre (ASC)
- Staff - HR

- ☐ Yes
- ☒ No

Sensitive Topics

Will your research involve sensitive topics such as:

- Criminal activity
- Traumatic experiences like those detailed above
- Self-identity i.e. gender, national, ethnic or racial identity
- Body image
- Mood or mental health conditions

☐ Yes

☒ No

Risks

Is there any significant risk (inc. physical/psychological harm or distress) to the researcher and / or any participants, field assistants, students, collaborators involved in the project?

☐ Yes

☒ No

Will your research involve the use of deception, the withholding of any information about the aims of the research or anything other than total transparency over your role as a researcher or what is involved in the research?

☐ Yes

☒ No

Data Risks

You may wish to read the guidance on [confidentiality and data protection](#), which includes definitions, before completing the next questions.

Will your research involve collection, creation, inference, or processing of **personal data**?

Personal data is information relating to natural living persons who can be identified directly from the information in question or who can be identified indirectly from that information in combination with other information.

If you are unsure if the data will be personal data, seek advice from Data Protection - dataprot@st-andrews.ac.uk

☒ Yes

☐ No

Will your research involve collection, creation, inference or processing of **special category data**?

Special category data is personal data relating to race, ethnic origin, politics, religion or philosophical beliefs, trade union membership, genetics, biometrics (where used for identification purposes), health, sex life or sexual orientation.

If you are unsure if the data will be special category data, seek advice from Data Protection - dataprot@st-andrews.ac.uk

☐ Yes

☒ No

Will your research involve audio-visual recordings or photographs of participants or members of the public?

☐ Yes

☒ No

Will your research involve collection, creation or inference of any other personal, confidential or sensitive data where you feel this might cause distress or that could cause harm should this data be intercepted?

☐ Yes

☒ No

Your responses to the filter questions indicate your application meets the criteria for standard review. Please click 'ok' to continue.

If you feel this review route is not appropriate, please review your answers to the filter questions or contact your School ethics committee for advice.

☒ Ok

Project Outline

Project title

User-Centred Design of Games for Gaelic Digital Normalisation

Project description (summary) (maximum 900 characters).

Give a concise narrative description without technical terminology of what you are proposing to do; who your participants are (e.g. age, organisation) and how they will be approached/ recruited; where the research will take place (e.g. site, country); what methods you will use, (e.g. survey, interview). (see [exemplars](#))

Note: the information you provide in this section will be reviewed by UTREC as part of ongoing monitoring activities. Your summary should therefore capture the key details of your application without requiring the reader to refer to or read other parts of the form.

This project will develop and test a digital game to support active Gaelic usage among pupils in Gaelic Medium Education (ages 8–12). Research will involve speaking with teachers and a supervised classroom session where pupils evaluate a prototype, supported by optional staff and pupil feedback through short questionnaires or informal interviews / focus groups.

Ethical considerations (summary) (maximum 900 characters)

Give an overview of both **the ethical issues raised** by your research and **how you will address** them (see [exemplars](#)). This could include: the risks and benefits, how you will ensure consent is voluntary and informed; confidentiality and how your data will be managed to protect this; potential risks to participants such as distress or reputational harm.

This should not substantially duplicate the information given in 'Project description' above.

Note: the information you provide in this section will be reviewed by UTREC as part of ongoing monitoring activities. Your summary should therefore capture the key details of your application without requiring the reader to refer to or read other parts of the form.

All participation will be voluntary, with parental/guardian consent mandatory and staff assistance for recruitment. Pupils and their parents/guardians may withdraw the pupil's participation at any time without consequence. Names and consent forms will be securely managed by myself. I will assign all of the pupils a participation code, which will then be used as identifiers on all information gathered on the day, ensuring that all data is pseudonymised at source. Participants (parents/guardians representing a pupil, or the pupil themselves) have 48 hours after the data is collected to withdraw from the research before their input is fully anonymised and transferred to a safe university cloud. Risks are minimised through a culturally familiar context, respectful facilitation, and clear research aims. Any concerns will be managed promptly, with well-being as a priority.

Estimated start date

Estimated end date

Research Funding

Is the research funded?

Is the research funded?

- ☐ Yes
- ☒ No

Internal Collaboration

Does the research involve collaboration with other researchers, Schools or Units, internal to the University?

- ☐ Yes
- ☒ No

External Collaboration

Does the research involve collaboration with researchers from other institutions or organisations, external to the University?

- ☒ Yes
- ☐ No

External Collaborator Details

Title

Dr

First Name

Will

Surname

Lamb

Email

W.Lamb@ed.ac.uk

Affiliation

2nd Supervisor

Role

Dr Will Lamb is a secondary advisory figure, fluent in Gaelic with expertise in Language Technology and Research. He will in no way be contributing directly to the research or carrying out any research as part of the project. I am making the Ethics Team aware of his position as a second supervisor.

Confirm there is a collaborative framework in place to ensure all collaborators and contributors are given appropriate recognition in any outputs

☒ I confirm

Research Setting

What is the geographic location(s) of the research - if you are conducting research remotely please indicate both your location and that of your participants or samples? If the location of participants will not be known, select 'no specific location'

- ☒ UK (exclusively)
- ☐ Other or multiple locations
- ☐ No specific location

If the research is being conducted in the UK please select relevant the devolved nations

- ☐ All
- ☒ Scotland
- ☐ England
- ☐ Wales
- ☐ Northern Ireland

Describe the setting of the research (this may be a physical context such as a type of place or a non-physical context such as an online environment).

You can repeat this question if you have multiple settings.

The site is Stornoway Primary / Bun-Sgoil Steornabhaigh, a Scottish Gaelic Medium Education primary school classroom where pupils are approaching fluency in the Gaelic language. The context is a physical environment: a familiar school classroom during regular hours with normal teacher supervision. Pupils will use electronic devices to participate in the game evaluation, followed by an oral discussion.

Have you permission to access the site of the research?

- ☒ Yes
- ☐ No
- ☐ Not Required

By whom (10 - 2,500 characters)

Dr Donald Weir, Gaelic Officer at Comhairle Nan Eilean Siar & Mrs Laura MacIver, Head Teacher at Bun-sgoil Steornabhaigh (Stornoway Primary School)

Evidence of permission (optional)

Documents

Type	Document Name	File Name	Version Date	Version	Size
Evidence of Permission Document	school-participation-agreement-email	school-participation-agreement-email.pdf			520.0 KB
Evidence of Permission Document	CNES-e-mail-confirmation	CNES-e-mail-confirmation.pdf			438.0 KB

Research Aims and Design

Summarise the aims of the research (max 6000 characters)

The purpose of this project is to investigate and prototype Human-Computer Interaction (HCI) strategies that allow active Gaelic language use among upper primary pupils in Gaelic-medium education (GME) through a purpose-built game. The central focus is to identify and evaluate interaction design principles that promote meaningful Gaelic use in digital environments, addressing the current gap in culturally and linguistically responsive technology for minority language learners.

Core objectives include:

Primary Objectives:

- Systematically reviewing cross-cultural HCI design frameworks and synthesizing context-sensitive strategies most relevant for Gaelic-medium settings.
- Identify and formalise design requirements for a Gaelic digital game, emphasising interface elements and interaction strategies that incentivise active Gaelic usage.
- Prototype a functional digital game that maximises opportunities for users to interact with Gaelic digitally.

Secondary Objectives

- Improve the prototype through teacher and pupil feedback, focusing on usability, cultural engagement, and the effectiveness of Gaelic language integration.
- Map and critically analyse game-based HCI principles against established curricular frameworks, identifying which design choices increase active Gaelic language engagement.
- Design in-game support systems (tutorials & feedback) that enable self-guided play, minimising the need for adult/supervisor support.

Tertiary Objectives:

- Document required adaptations for implementation to future applications.
- Investigate applicability to other minority languages.
- Explore teacher-facing configurations (admin interface).

The research will use a user-centred, mixed-methods research approach. The process follows several stages:

1. Literature Review:

- Conduct a structured literature review of Gaelic and the GME user group. Then, explore areas of HCI and game design that are relevant to minority-language / cultural game design for children.
- Synthesise key cross-cultural and language-specific design principles relevant to GME.
- Identify limitations and needs unique to Gaelic digital education.

2. Design Requirements Gathering:

- Formalise user requirements and language-specific interface considerations. Special attention will be given to age-appropriateness, linguistic complexity, and mechanisms for active language use.

3. Prototype Development:

- Construct a usable game that integrates Scottish Gaelic with its culture appropriately.

4. Evaluation:

- By deploying the prototype in GME primary classroom settings (with appropriate consent, ethics approval, and teacher collaboration), mixed methods will be used to obtain data.
- Quantitatively: Metrics will be gathered from in-game performance data, and a multiple-choice digital feedback form will be presented to children following gameplay.
- Qualitatively: During gameplay, observational fieldnotes will be taken to record any issues faced by participants. Two open-ended questions will be included at the end of the questionnaire for children to reflect on their gameplay experience. Teachers will be asked to complete a questionnaire, and a follow-up interview may be conducted if necessary. A discussion group will take place after the gameplay, where field notes will also be taken.

5. Analysis and Refinement:

- Analyse evaluation data to refine interface elements, language prompts, and game design.
- Record rationales for major design changes with reference to HCI theory and observed user needs.

6. Data Management and Ethics:

- All participation is voluntary and anonymised, with parental consent assisted via the school, which will manage the forms and give them to me for signing via initial visit or email.
- Researcher access is limited to anonymised participant numbers, protecting pupil identity and confidentiality.
- No sensitive or identifiable personal data will be stored by the researcher (see ethics section).
- Short mock session, ensuring briefing and activity suitability by pre-testing the game with the school teachers before the research session. The game's content should be verified for age and culture appropriateness.

7. Reporting and Outcome:

- Map the effectiveness of game and HCI design choices to established curricular goals and language engagement indicators.
- Produce a final dissertation, recommending best practices for minority-language digital HCI.

Participants

This section asks about your research participants.

Use 'add another' to repeat this section if you have more than one participant group or set.

Give this participant group a name or label, for example 'Sample 1', 'Fife group', 'Qualitative dataset'

Primary GME Pupils - Core User Evaluation

Lower Age Limit (years)

8

Upper Age Limit (years)

12

Inclusion Criteria

- Currently enrolled in P5-P7 within a Scottish Gaelic-medium education setting
- Functionally literate in Scottish Gaelic (able to read and understand age-appropriate Gaelic text)
- Regular classroom technology use experience (basic computer/tablet familiarity)
- Parental/guardian consent provided via school administration
- Pupil consent obtained on day of participation

Exclusion Criteria

- Pupils whose parents/guardians have not provided written consent
- Pupils who demonstrate distress or unwillingness to participate on the day of testing
- Non-GME pupils or those with insufficient Gaelic comprehension for meaningful interaction

How many participants do you intend to recruit or expect to participate?

If you're not sure, please provide an estimate. If you have more than one group, please provide the number for this group rather than the total.

35

Summary (optional)

Use this space if you wish to give a summary or overview of your participants or provide any additional information not already included. (max 6000 characters)

Given the small population of GME pupils and specialised nature of this research, this represents a robust sample for feedback. The age range (P5-P7, ages 8-12) corresponds to pupils who have developed sufficient Gaelic literacy for meaningful digital interaction. Note, there is a possibility, depending on the school class setup, this could be composed of two groups i.e P5/6 class and a P7 class.

Use 'add another' to repeat this section if you have more than one participant group or set.

Give this participant group a name or label, for example 'Sample 1', 'Fife group', 'Qualitative dataset'

GME Teaching Staff

Lower Age Limit (years)

22

Upper Age Limit (years)

100

Inclusion Criteria

Currently teaching within GME settings
Professional experience with Gaelic-medium pedagogy
Familiarity with educational technology integration

Exclusion Criteria

N/A

How many participants do you intend to recruit or expect to participate?

If you're not sure, please provide an estimate. If you have more than one group, please provide the number for this group rather than the total.

5

Summary (optional)

Use this space if you wish to give a summary or overview of your participants or provide any additional information not already included. (max 6000 characters)

This secondary group provides an expert perspective on cultural authenticity, curricular alignment, and classroom integration potential, following established mixed-method evaluation approaches in educational HCI research.

Informed Consent

This section asks questions related to recruitment and participation, including around informed consent and factors which can impact this.

Describe how you will identify and approach potential participants (max 10000 characters)

Comhairle nan Eilean Siar identified the school as the most suitable site for the reason that it is the biggest GME primary school in the area. My targeted age group of P5-7 narrows down the participants to a pre-existing group, where the opportunity will be offered to all. Information sheets will be given to both parents and teachers. I have spoken with the head teacher over the phone to ensure participant recruitment will be unbiased, non-tested, and open to all.

Upload any materials your will be using to recruit participants such as adverts, letters to schools / parents/ guardians

Documents					
Type	Document Name	File Name	Version Date	Version	Size
Recruitment Document	letter-to-parentguardian	letter-to-parentguardian.pdf	30/11/2025	1.2	106.8 KB
Recruitment Document	letter-to-school-in-principle	letter-to-school-in-principle.pdf	30/11/2025	1.2	113.7 KB
Recruitment Document	letter-to-school-commencing	letter-to-school-commencing.pdf	30/11/2025	1.2	126.6 KB

Will you be giving participants information about the research so they can make an informed decision whether to participate?

- ☒ Yes
- ☐ No

Upload any participant information materials you will be using, such as participant information sheets

Documents

Type	Document Name	File Name	Version Date	Version	Size
PIS Document	PG_PIS_[NOV 2025]_[V1.2]_[USER-CENTRED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION]	PG_PIS_[NOV 2025]_[V1.2]_[USER-CENTRED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION].pdf	30/11/2025	1.2	183.2 KB
PIS Document	Child Participant Information Script	Child Participant Information Script.pdf	30/11/2025	1.0	240.9 KB
PIS Document	T_PIS_[NOV 2025]_[V1.2]_[USER-CENTRED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION]	T_PIS_[NOV 2025]_[V1.2]_[USER-CENTRED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION].pdf	30/11/2025	1.2	150.5 KB

Will you be obtaining explicit consent from individual participants?

- ☒ Yes
- ☐ No

Upload any consent materials you will be using, such as consent forms

Documents

Type	Document Name	File Name	Version Date	Version	Size
Participant Consent Document	Child Participant Information Script	Child Participant Information Script.pdf	30/11/2025	1	240.9 KB
Participant Consent Document	T_Consent form_[NOV 2025]_[V1.2]_[USER-CENTRED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION]	T_Consent form_[NOV 2025]_[V1.2]_[USER-CENTRED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION].pdf	30/11/2025	1.2	126.9 KB
Participant Consent Document	PG_Consent form_[NOV 2025]_[V1.2]_[USER-CENTRED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION]	PG_Consent form_[NOV 2025]_[V1.2]_[USER-CENTRED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION].pdf	30/11/2025	1.2	120.9 KB

Describe your consent process (max 6000 characters)

Phase 1: Information & Initial Consent (Pre-Research)

Parent/Guardian Consent:

Written consent is obtained from parents/guardians through a detailed Participant Information Sheet (PIS) the 0-12 child PIS form template was used along with the advised formal parental/guardian consent form. The process begins when information sheets are distributed by school staff alongside consent forms. Parents/Guardians receive comprehensive details about the study, their child/ward's participation, potential risks (assessed as minimal), data protection measures, and their child/ward's right to withdraw. Importantly, the consent materials explicitly state that signing the form does not commit their child/ward to anything they do not wish to do. This multi-layered approach ensures parents/guardians understand participation is entirely voluntary.

Phase 2: Child Assent (Day of Research)

Age-Appropriate Explanation:

At the beginning of the research session, I will explain the study to each child in clear, age-appropriate language. This verbal explanation confirms their willingness to participate and ensures they understand what they are consenting to, rather than relying solely on parental signatures. I will read out the script as written in the Child Participant Information Script.

Ongoing Assent & Withdrawal:

Children are reminded throughout the session that they can stop participating at any time without explanation or consequence. This ongoing assent recognises that written consent alone is insufficient when working with younger participants. I will monitor body language and provide regular verbal check-ins to gauge comfort levels. Please refer to section 2 of the Child Participant Information Script to see the states of assent for a child.

Immediate Cessation Protocol:

If any child shows signs of distress, discomfort, or unwillingness, research activities will be stopped immediately. This safeguard prioritises participant well-being over data collection efficiency.

Phase 3: Activity-Specific Consent

My consent forms employ a modular consent approach, allowing parents to consent to specific components of the research:

Game-playing activities (30-40 minutes)

Observational data collection (written observations + discussion group)

Screen recording (with explicit note that no personal identifiers will be captured)

Questionnaire (age-appropriate multiple-choice and open-ended questions)

This granular approach gives parents/guardians and children greater control over which aspects of the research they feel comfortable with, rather than an all-or-nothing consent model.

Teacher Consent

Teachers undergo a separate, parallel consent process through their own PIS and consent form. Teachers consent to:

- Supervising pupils during the research session and providing feedback
- Offering professional reflections on game appropriateness
- Participating in a brief post-session questionnaire
- Allowing anonymised quotes from their professional reflections to be used in research outputs

Teachers explicitly understand they are empowered to halt or intervene in any research activity if they have safeguarding or professional concerns, they are not passive observers but active safeguarding participants.

During the recruitment, information, or consent process will you tell participants that their participation is voluntary and that they can decline to participate with no disbenefit?

- ☒ Yes
- ☐ No

During the recruitment, information, or consent process will you tell participants that they may withdraw from the research within the time specified in the PIS and for any reason, without having to give an explanation, and with no disbenefit?

- ☒ Yes
- ☐ No

During the recruitment, information, or consent process will you tell participants that their data will be treated with full confidentiality?

- ☒ Yes
- ☐ No

During the recruitment, information, or consent process will you tell participants that if published or shared, the data will not be identifiable as theirs?

- ☒ Yes
- ☐ No

During the recruitment, information, or consent process will you tell participants how their data will be stored, who will have access to it, and when the data will be destroyed?

- ☒ Yes
- ☐ No

Will participants be able to continue to participate if they reject or omit parts of the study? For example, declining to be recorded, omitting survey questions, declining to give a certain sample

- ☒ Yes
- ☐ No

Will any financial inducement be offered?

- ☐ Yes
- ☒ No

Are any of the participants in a dependent relationship with the investigator?

- ☐ Yes
- ☒ No

Will you be debriefing or giving 'end of study' information to participants?

- ☒ Yes
- ☐ No

Use the upload function to attach any written debrief document or protocol. Otherwise, describe any debrief protocol or process you will be using (max 6000 characters)

Following the session(s), participants will receive age-appropriate explanations of what they contributed to the research and how their feedback helps improve digital resources for Gaelic learners. Please refer to the Child Participant Information Script, section 3, for the debriefing routine.

The debriefing will include thanking participants for their contributions, explaining how their gameplay and feedback inform design improvements, and sharing that their participation helps create better digital tools for GME. This approach reinforces the research community's benefit rather than an extractive purpose.

Teachers will also receive feedback about prototype performance and participant responses, enabling them to provide context and follow-up discussions if necessary. This extends the educational value beyond the research session itself.

Written summary reports will be provided to schools which then may be distributed to parents/guardians and their pupils, outlining general findings and their implications for GME digital resources. This transparent debriefing honours participants' contributions while maintaining anonymity.

Upload your debrief document or protocol (optional)

Type	Document Name	Documents		Version Date	Version	Size
		File Name				
Debrief Document or Protocol	T_Debrief-Letter	T_Debrief-Letter.pdf		30/11/2025	1.0	104.1 KB
Debrief Document or Protocol	PG_Debrief_Letter	PG_Debrief_Letter.pdf		30/11/2025	1.0	103.0 KB

Will you be using a distress protocol? This is optional but you may wish to consider this if your research is likely to touch on distressing topics.

- ☐ Yes
- ☒ No

Research Activities

This section asks about any research activities participants will be undertaking. This helps reviewers understand the potential impact or burden on participants.

Will participants be asked to undertake an activity for the purposes of the research?

'By 'activity' we mean any active engagement with the research. For example, this could include participating in an interview, completing a questionnaire, partaking in tests or measurements, or undertaking a task.'

- ☒ Yes
- ☐ No

Activity Details

What will participants be asked to do? (max 6000 characters)

Participants will engage in structured gameplay sessions with a prototype Gaelic game designed specifically for this research. The activities are designed to be interactive and enjoyable rather than extractive or clinical.

Specific Tasks May Include:

- Navigating game interfaces using touch or mouse interaction
- Reading and responding to Gaelic prompts and instructions
- Solving a game problem that may require strategic thinking
- Making choices within scenarios presented in Gaelic
- Completing a game that reinforces language use

Following gameplay, participants will provide simple feedback through a multiple-choice questionnaire and 2 open-ended questions.

Estimate duration per participant (use an appropriate metric such as minutes or hours)

45-60 minutes

Estimated number of occurrences per participant

1

Time between occurrences (use an appropriate metric such as minutes, hours or days)

N/A

List any measures, instruments, or equipment you will be using. (max 10000 characters)

Tablet / Laptop device to take notes. (typed, not dictated). Tablet / Laptop will be used for children to test the prototype - provided by the school.

Upload copies of questionnaires, interview schedules, focus group guide, experiment protocols etc.

Documents

Type	Document Name	File Name	Version Date	Version	Size
Default	Discussion Group Guide	Discussion Group Guide.pdf	30/11/2025	1.0	101.5 KB

Will you be collecting data any other way i.e. observation, passive data collection

- ☒ Yes
- ☐ No

Describe (max 6000 characters)

Observational data will be collected during gameplay sessions, focusing on usability behaviors, engagement indicators, and natural language use patterns. This observational approach follows established protocols for educational HCI research with children.

Data Collection

You may find the guidance on [Confidentiality and data protection](#), which includes definitions and relevant University contacts, useful to answer these questions. You should aim to collect the minimum amount of data necessary for your research (see 'data minimisation').

If you need help to think through or plan how you will manage your research data then contact the University's [Research Data Management](#) team (research-data@st-andrews) for advice or to develop a Data Management Plan.

For advice on personal data or GDPR, contact [Data Protection](#) (data-prot@st-andrews.ac.uk).

For help with IT equipment, technology or processes, contact [IT Services](#) (itservicedesk@st-andrews.ac.uk)

You may find the guidance on [Confidentiality and data protection](#), which includes definitions and relevant University contacts, useful to answer these questions.

How will you collect data? (this means the technology/software/hardware/paper etc. rather than specific questionnaires or measures)

- ☒ Qualtrics, MS Forms
- ☐ IT-approved encrypted recording device
- ☒ Other (describe below)
- ☐ Not applicable

Specify Other (max 6000 characters)

I will be screen recording gameplay (screen only), a digital questionnaire after the game, as well as field note taking.

Digital questionnaires will be administered using Qualtrics.

Qualtrics Form for Pupils:

https://standrews.eu.qualtrics.com/jfe/form/SV_eszZmcK9LrP6Rg2

Qualtrics Form for Teaching Staff:

https://standrews.eu.qualtrics.com/jfe/form/SV_2IAUTFvdgpNaqNM

Both forms ranked 'Great' by Qualtrics. Should you wish to fill out the forms with redundant information to trial them, I welcome doing so, and I can delete those responses before the experiment.

Have you checked this means of data collection is acceptable?

If you are working with personal data you must check with [Data Protection](#).

If you have any concerns about cybersecurity or technology, check with [IT Services](#).

For help with and planning for research data management, speak to the [Research Data Management](#) team.

- ☒ Yes
- ☐ No

In what form will the data be collected?

- ☐ Anonymous
- ☒ Pseudonymous
- ☐ Identifiable
- ☐ Mix (Please describe below)

Data Transfer

You may find the guidance on [Confidentiality and data protection](#), which includes definitions and relevant University contacts, useful to answer these questions.

Will you need to transfer the data?

- ☒ Yes
- ☐ No

In what form will the data be transferred?

- ☒ Anonymous
- ☐ Pseudonymous
- ☐ Identifiable
- ☐ Mix (Please describe below)

How will you securely transfer the data? (max 6000 characters)

Primary Transfer Requirements:

From school to myself, the researcher: Names and personally identifiable information will be transferred from the school to the researcher. The school will retain all identifiable information (names, consent forms) and provide numerical information for participants who have parental / guardian and individual consent until I visit the site to collect and sign the consent forms. From then, any data collection will map the identity of a participant to a number, meaning all data is pseudonymised at source. This protects participant identity while enabling data collection.

From Data Collection Devices to University Systems: Observational notes, feedback responses, and technical metrics will be anonymised, then transferred from collection devices (tablet / laptop / observation sheets) to secure University of St Andrews data storage systems within 48 hours of collection.

Transfer Security Measures:

All transfers will use password-protected documents

Digital transfers will occur via University networks or approved cloud services

All transfer activities will be logged and documented

Data Minimisation:

Only essential research data will be transferred. No personal identifiers, school-specific information beyond general location (e.g., "CNES / Stornoway Primary School"), or sensitive personal information will be transferred. The separation of identity management from research data ensures minimal data exposure during transfer.

I have contacted dataprot@st-andrews.ac.uk and liaised with Christopher within that department to ensure my actions as the data controller minimizes personal / identifiable data, and to ensure that my practice is compliant with both University and legal guidance.

Compliance:

This approach aligns with University data protection policies, maintains GDPR compliance, and follows best practices for educational research with child participants. However, I should note that any data breach or loss during transfer will be immediately reported to dataprot@st-andrews.ac.uk.

Data Storage

You may find the guidance on [Confidentiality and data protection](#), which includes definitions and relevant University contacts, useful to answer these questions.

How will you store the data?

- ☒ University network drive or OneDrive/Teams
- ☐ Other (describe)

In what form will the data be stored?

- ☒ Anonymous
- ☐ Pseudonymous
- ☐ Identifiable
- ☐ Mix (Please describe below)

Data Access

You may find the guidance on [Confidentiality and data protection](#), which includes definitions and relevant University contacts, useful to answer these questions.

Who will have access to the data during the study?

- ☒ Researcher and supervisor
- ☐ Others (describe)

Will there be any access controls?

- ☒ University Network Drive or OneDrive/Teams controls – specified users, accessed via SSO
- ☐ None
- ☐ Other (describe)

Data Form Changes

You may find the guidance on [Confidentiality and data protection](#), which includes definitions and relevant University contacts, useful to answer these questions.

Do you plan to pseudonymise or anonymise the data at some point during the study? For example, converting the data from identifiable to anonymised by removing identifiers or from pseudonymised to anonymised by deleting a pseudonymisation key.

- ☒ Yes
- ☐ No

Describe your approach and any key time points (max 6000 characters)

I will pseudonymise data before any research takes place, by assigning children numbers. Those numbers will then be held attached to the data for 48 hours. After that time, full anonymisation will take place when data is transferred onto the secure cloud (St Andrews OneDrive).

Data Sharing

You may find the guidance on [Confidentiality and data protection](#), which includes definitions and relevant University contacts, useful to answer these questions.

Will you be sharing the data?

- ☒ Yes
- ☐ No

In what form will the data be shared?

- ☒ Anonymous
- ☐ Pseudonymous
- ☐ Identifiable
- ☐ Mix (please describe)

Where will the data be shared? (max 6000 characters)

Academic Dissertation: Anonymised findings, quotes, and insights will be shared in the final dissertation as required for the CS4099 Major Software Project. All data will be completely anonymised with no possibility of participant identification

Will the data be

- ☒ Publicly available
- ☐ Access controlled
- ☐ Shared with specific people
- ☒ Other (describe)

Please describe (max 6000 characters)

If paper is published, the anonymised data and findings may be available.

Will there be any constraints on who can access the shared data?

- ☐ Yes
- ☒ No

Data Publication

You may find the guidance on [Confidentiality and data protection](#), which includes definitions and relevant University contacts, useful to answer these questions.

Will you be publishing the data?

- ☒ Yes
- ☐ No

In what form will the data be published?

- ☒ Anonymous
- ☐ Pseudonymous
- ☐ Identifiable
- ☐ Mix (Please describe)

Where will the data be published?

- ☒ Dissertation or thesis
- ☒ Scholarly outputs such as academic journal articles or conference presentations
- ☐ Other (describe)

Will you share the results with participants?

- ☒ Yes
- ☐ No

Describe how you will share your results with participants (max 6000 characters)

Following the research session, participants will receive immediate, age-appropriate verbal feedback about their contributions. This includes thanking them for their participation, explaining how their gameplay and feedback help improve digital resources for Gaelic learners, and highlighting that their involvement contributes to better educational tools for their community. Within 4-6 weeks of data collection completion, the participating school will receive a written summary report outlining the general research findings and key insights. The teacher may then share this information with the students in the class.

Data Retention

You may find the guidance on [Confidentiality and data protection](#), which includes definitions and relevant University contacts, useful to answer these questions.

This section asks questions about retention of **locally held** data beyond the end of the study i.e. data which is held by the research team, such as on a network drive, and not any data which has been formally shared, deposited or archived.

Will you be retaining any locally held data beyond the end of the study?

- ☐ Yes
- ☒ No

Data Destruction

Will the data be securely destroyed following the study and/or the retention period?

- ☒ Yes
- ☐ No

Data Management Plan

Do you have a Data Management Plan you wish to include with this application?

- ☐ Yes
- ☒ No

Data - Participant info

Does the information you have provided in the preceding data management questions reflect that which will be given to participants, for example in participant information or consent forms?

- ☒ Yes
- ☐ No

Risk-Benefit Balance

Please provide a clear, concise description of the anticipated benefits of the research. This could be to the participant, the participant's community, the academic community, wider society, or in developing your own research skills. (max 10000 characters)

Direct Benefits to Child Participants

Participants will benefit through engaging with a specially designed game that celebrates their linguistic heritage. Research demonstrates that game-based learning produces moderate to large effects on cognitive development, social skills, emotional regulation, motivation, and engagement in primary-aged children.

The game design prioritises authentic Gaelic language use, providing participants with rare opportunities to experience their heritage language as the natural medium for digital interaction. This addresses a critical gap identified in GME, where pupils typically encounter English-dominant technology that undermines language immersion and cultural belonging. By positioning Gaelic as the default interface language, participants gain confidence in digital Gaelic literacy while developing technological competencies essential for modern education.

Community and Cultural Benefits

This research directly addresses urgent needs within the GME community, where digital resources remain severely limited beyond basic literacy tools. The design process ensures that community voices shape the design, creating genuine ownership and relevance.

The research contributes to broader Gaelic language preservation efforts by demonstrating how technology can support rather than undermine minority language education. By creating successful models for Gaelic digital engagement, this work provides frameworks for other minority language communities facing similar challenges with digital inclusion and cultural preservation.

Academic and Research Community Benefits

This study advances the limited but crucial field of HCI research for Gaelic, contributing novel insights about cross-cultural interface design, user experience considerations for child learners, and effective integration strategies for heritage languages in educational technology. The research fills significant gaps in understanding the HCI needs for this user group.

Societal and Policy Benefits

The research could generate evidence to inform educational policy decisions about digital resource allocation for GME settings. By demonstrating effective approaches to minority language technology integration, findings can influence funding priorities or curriculum development. This contributes to Scotland's National Gaelic Language Plan objectives of increasing Gaelic use across diverse contexts, including digital environments.

Personal and Professional Development Benefits

Selected by the Global Office for a year abroad in Hong Kong, seamlessly managing three languages across education, culture, and media - I witnessed firsthand the power of multilingual systems in action. Supporting minority languages in education is not an abstract idea, it is a lived reality. From that moment, my mission evolved, to bring home proven models and adapt them to Scotland, driving real change so that Gaelic and other minority language users are empowered, not marginalised, by our educational technology.

As a former GME pupil conducting this research, I bring authentic cultural perspective while developing advanced research skills in HCI, game design, and minority language pedagogy. The interdisciplinary nature of the project enhances my own expertise across computer science, linguistics, education, and cultural studies - preparing for future contributions to both academic and applied work in these fields.

Long-term Impact Potential

Beyond immediate research outcomes, this work establishes foundations for sustained GME technology development. The game prototype serves as a proof-of-concept for larger-scale projects, while documented design principles enable scaling to other educational contexts and minority language communities. The research creates pathways for future collaborations between technology developers, educators, and cultural organisations focused on digital language preservation.

Gaelic Medium Education is growing across Scotland's urban and rural landscapes year on year, representing one of the most vital forces in cultural and linguistic continuity. Yet as classrooms become more digital around the globe, the prevailing assumption that a single technological solution can serve all learners and contexts is both flawed and dangerous. To impose uniform digital tools without sensitivity to cultural and linguistic realities is not merely an inconvenience - it risks eroding heritage, normalising linguistic marginalisation, and forcing entire communities to "make do" with systems that quietly disadvantage them. The stakes extend beyond academia to questions of equity, cultural survival, and the right to a fully modern education without surrendering one's language.

Considering any residual risks indicate why you believe there is a favourable risk-benefit balance. (max 10000 characters)

After comprehensive risk mitigation measures, the residual risks to participants are minimal and temporary. The primary remaining risks include:

- The game may not be suitable for children's age and stage of development
- The game may not be culturally appropriate
- Minor frustration during gameplay (mitigated by adaptive difficulty and teacher support)
- Brief fatigue from 45-60 minute sessions (can be addressed through flexible timing and break options)
- Technical issues (managed via backup equipment, and researcher's technical expertise)
- Minimal data protection concerns i.e child names used in classroom (eliminated through complete anonymisation at source, no such information will be stored or used in any research avenue)

Temporary, minor inconveniences cannot reasonably outweigh substantial educational, cultural, and research benefits.

Ethical Issues

Please provide a clear, concise description of the ethical issues raised and how you will address them (see exemplars). (max 10000 characters)

1. Game Appropriateness

Before conducting the main research session, the researcher will conduct a brief mock session (30 minutes) with the classroom teacher(s) who will supervise the research activity. This will include:

- Demonstration of the game prototype and research activities
- Teacher feedback on age-appropriateness and cultural sensitivity
- Verification that the briefing, game content, and evaluation activities are suitable
- Confirmation of timing and technical requirements
- Assessment of any safeguarding considerations
- Opportunity to suggest modifications before the main session

This ensures full alignment with school expectations.

2. Consent and Capacity Issues

Issue: Children aged 8-12 have developing capacity to understand research participation, requiring careful attention to both child assent and parental consent processes.

How Addressed: A multi-layered consent framework will ensure both voluntary participation and appropriate protection.

Parents/guardians will receive comprehensive information sheets and provide written consent. Schools will facilitate this process, ensuring participants are well informed and happy to participate. On the research day, children will receive age-appropriate verbal explanations of activities and provide verbal assent. Ongoing consent will be monitored throughout, with immediate withdrawal options clearly communicated. The research design respects children's developing autonomy while ensuring appropriate adult oversight.

3. Power Imbalances and Voluntary Participation

Issue: The inherent power differential between adult researchers and child participants may compromise genuinely voluntary participation, especially in school settings where children are accustomed to adult authority.

How Addressed: Multiple safeguards will minimise coercive influences. Teachers will emphasise that participation is completely optional and unrelated to academic performance. The game-based methodology creates enjoyable rather than obligatory experiences. Children can withdraw at any time without explanation or consequence. My researcher's role will be presented as facilitator rather than an authority figure, with teachers present to provide familiar support. There will be a small stationery item and a certificate given to participating students; however, this is in no way used as a strong incentive to participate and will not be announced/given until the session has been completed.

4. Data Protection and Privacy

Issue: Children have heightened privacy rights requiring exceptional data protection measures, particularly in minority language contexts where community identification risks are elevated.

How Addressed: I will pseudonymise data at source, eliminating privacy risks - no personal identifiers will be collected during research. The school and I, the researcher, will maintain consent documentation separately from research data, and I will only use numerical data to identify participants. All data will be stored securely using University-approved systems with restricted access. Published

findings will use anonymised results only, with no individual responses identifiable. The separation of identity from research data provides robust protection.

5. Cultural Sensitivity and Linguistic Rights

Issue: Research involving minority language communities carries risks of cultural misrepresentation or exploitation, requiring exceptional sensitivity to Gaelic educational contexts and community values.

How Addressed: My own extensive GME background and lived experience provides an authentic cultural perspective. All activities will celebrate rather than study Gaelic language use, positioning the research as community-serving rather than extractive. Consultation with GME educators ensures cultural appropriateness throughout. The research aims to strengthen rather than analyse Gaelic education, with findings shared to benefit the broader GME community. Materials and interactions will reflect deep respect for Gaelic linguistic heritage.

6. Risk of Distress or Harm

Issue: While minimal, research activities could potentially cause frustration, anxiety, or discomfort for some participants, requiring proactive risk management.

How Addressed: The game-based design prioritises enjoyment and age-appropriate challenge levels. Sessions occur in familiar school environments with teacher supervision providing immediate support. The brief duration (45-60 minutes) minimises fatigue while maintaining engagement.

Any signs of discomfort will result in immediate activity modification or cessation. Teachers familiar with each child's needs will monitor well-being throughout. The voluntary framework ensures that only comfortable, willing children participate.

7. Educational Disruption and Academic Impact

Issue: Research activities must not negatively impact children's educational experiences or create academic disadvantages for participants or non-participants.

How Addressed: Timing will be coordinated with teachers to minimise curriculum disruption, potentially integrating activities into existing curricular lessons. Participation decisions will have no impact on academic assessment or teacher relationships. The research design treats educational benefit as an equal with data collection.

8. Access Issues

Issue: Multiple gatekeepers (schools, teachers, parents) may create complex approval processes that could inadvertently exclude certain children or create selection bias.

How Addressed: Transparent communication with all stakeholders will ensure informed decision-making. Schools will be encouraged to offer participation to all eligible children without prejudgment. The voluntary framework respects all user decisions while maintaining inclusive access where consent is provided.

9. Long-term Impact and Data Use

Issue: Children may not fully understand how their participation contributes to long-term research outcomes or potential future uses of findings.

How Addressed: Age-appropriate debriefing straight after participation will explain how their participation helps improve digital resources for Gaelic users. Results will be shared with schools, demonstrating concrete benefits from their contributions. The research commitment to anonymised, educational use only provides clear boundaries for data application. Future research using these findings, quoting this work, should be informed to maintain the same ethical standards and community benefit focus.

End Questions

Describe here anything you would like to include that has not been covered so far (max 6000 characters)

Attached is guidance from Dr Dritschel, as well as Qualtrics questionnaires. I would encourage you to read the ethics phases of significance document, which orchestrates all of the documentation included in this ethics application in a sequence of critical phases.

Are there any other documents you wish to upload with your application?

- ☒ Yes
- ☐ No

Upload here

Documents					
Type	Document Name	File Name	Version Date	Version	Size
Application Supporting Document	PVG Guidance	PVG Guidance.png			148.5 KB
Application Supporting Document	Pupil Qualtrics Questionnaire Questions	Pupil Qualtrics Questionnaire Questions.pdf	30/11/2025	1.0	198.0 KB
Application Supporting Document	Teaching Staff Qualtrics Questionnaire Questions	Teaching Staff Qualtrics Questionnaire Questions.pdf	30/11/2025	1.0	55.4 KB
Application Supporting Document	Ethics-Phases-of-Significance	Ethics-Phases-of-Significance.pdf	01/12/2025	1.0	106.8 KB

At the start of this form you were asked to give a brief project description and summary of ethical considerations. You may wish to check you are happy with these now you have completed the rest of the form. Use the 'Navigate' function.

Declarations

I am aware of, understand and will enact my responsibilities as a researcher as detailed in:

The University's [Principles of Good Research Conduct policy](#) and [ethical guidelines](#)

☒ I confirm

Any relevant professional guidelines (e.g. BPS, MRC, ASA)

☒ I confirm

The University's Policy and guidance on [Data Management and Protection](#)

☒ I confirm

I am aware of the conditions of any funding associated with my work and will ensure that information given to my research participants is in line with those conditions.

☒ I confirm

I understand that I must store the final completed copy of this form as part of my research project paperwork.

☒ I confirm

Authorisation

Applicant authorisation (signature)

Signed: This form was signed by Luke Macleod (lcm31@st-andrews.ac.uk) on 10/12/2025 12:16

Signatures

Supervisor comments

Hi Luke, this looks in good shape to be submitted.

Supervisor declaration - I confirm that I have discussed the ethical implications of this project with the student applicant, that I have read this application, and that I approve its submission to the ethics committee for consideration

Signed: This form was signed by Dr Abd Alsattar Ardati (aaa8@st-andrews.ac.uk) on 10/12/2025 12:24

Involves children?

Involves children?

Yes